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YENEPOYA (DEEMED TO BE UNIVERSITY)

Yenepoya Institute of

Arts, Science, Commerce

and Management

Project – High Level Design
on

PREDICTIVE MODELLING FOR HEALTH CARE SUCCESS

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Table of Contents

1. Introduction. 6
   1.1. Scope of the document. 6
   1.2. Intended Audience 6
   1.3. System overview. 6
2. System Design. 6
   2.1. Application Design 6
   2.2. Process Flow. 6
   2.3. Information Flow. 7
   2.4. Components Design 8
   2.5. Key Design Considerations 8
   2.6. API Catalogue. 8
3. Data Design. 8
   3.1. Data Model 8
   3.2. Data Access Mechanism 9
   3.3. Data Retention Policies 9
   3.4. Data Migration 9
4. Interfaces 9
5. State and Session Management 10
6. Caching 10
7. Non-Functional Requirements 10
   7.1. Security Aspects 10
   7.2. Performance Aspects 10
8. References 11

1. Introduction

Stroke is a critical medical condition that occurs when the blood supply to the brain is interrupted or reduced, leading to deprivation of oxygen and nutrients. This results in rapid damage to brain cells and may cause permanent disability or death. According to global health studies, stroke remains one of the leading causes of mortality and long-term neurological impairment.

Traditional stroke diagnosis relies heavily on clinical observations, patient history, and medical imaging. While effective, these methods may not always identify hidden patterns in large datasets. With the advancement of machine learning, it is now possible to analyze complex healthcare data and extract meaningful insights that support early diagnosis.

The **Stroke Prediction using Machine Learning** system is designed to predict the likelihood of stroke based on patient health data. By analyzing attributes such as age, hypertension, heart disease, body mass index (BMI), glucose levels, and lifestyle factors, the system provides a predictive output that can assist healthcare professionals in making informed decisions.

The primary objective of this system is to enhance preventive healthcare by identifying high-risk individuals at an early stage and enabling timely intervention.

2. Scope of the Document

This document defines the high-level design of the stroke prediction system, focusing on architectural structure, data flow, and system components.

In Scope:

- Overall system architecture and layered design
- Data processing workflow and pipeline
- Functional modules and their responsibilities
- Data storage and access mechanisms
- Interface design and output representation
- Performance and security considerations

Out of Scope:

- Detailed code implementation
- Algorithm-level optimizations
- Low-level system configurations

3. Intended Audience

This document is intended for:

- **Academic Evaluators** – To assess the completeness and quality of system design
- **Faculty Mentors** – To review architecture and methodology
- **Developers** – To understand system structure and workflow
- **Students** – For academic learning and reference

4. System Overview

The system is designed as a **machine learning-based predictive system** that processes structured healthcare data to determine stroke risk.

Input Parameters:

- Age (numerical)
- Gender (categorical)
- Hypertension (binary)
- Heart Disease (binary)
- Body Mass Index (BMI)
- Average Glucose Level
- Smoking Status (categorical)

Output:

- **0 → No Stroke Risk**
- **1 → Stroke Risk**

The system processes input data through preprocessing, model training, and prediction stages to generate accurate outputs.

5. System Design

5.1 Application Design

The system follows a **layered 4-tier architecture**, ensuring modularity, flexibility, and scalability.

Presentation Layer

The Presentation Layer is responsible for user interaction with the system. It provides an interface for users to enter patient details and view prediction results. The layer ensures that outputs are displayed in a clear and user-friendly manner, often including basic visualizations and summaries to improve understanding.

Application Layer

The Application Layer acts as the core control unit of the system. It handles user requests, manages the overall workflow, and ensures proper communication between different layers. This layer performs input validation, controls data flow, and coordinates with the machine learning engine to generate predictions efficiently.

ML Engine Layer

This layer is the core of the system. It handles data preprocessing, feature engineering, model training, and prediction. The Decision Tree algorithm is used to classify whether a patient is at risk of stroke.

Data Layer

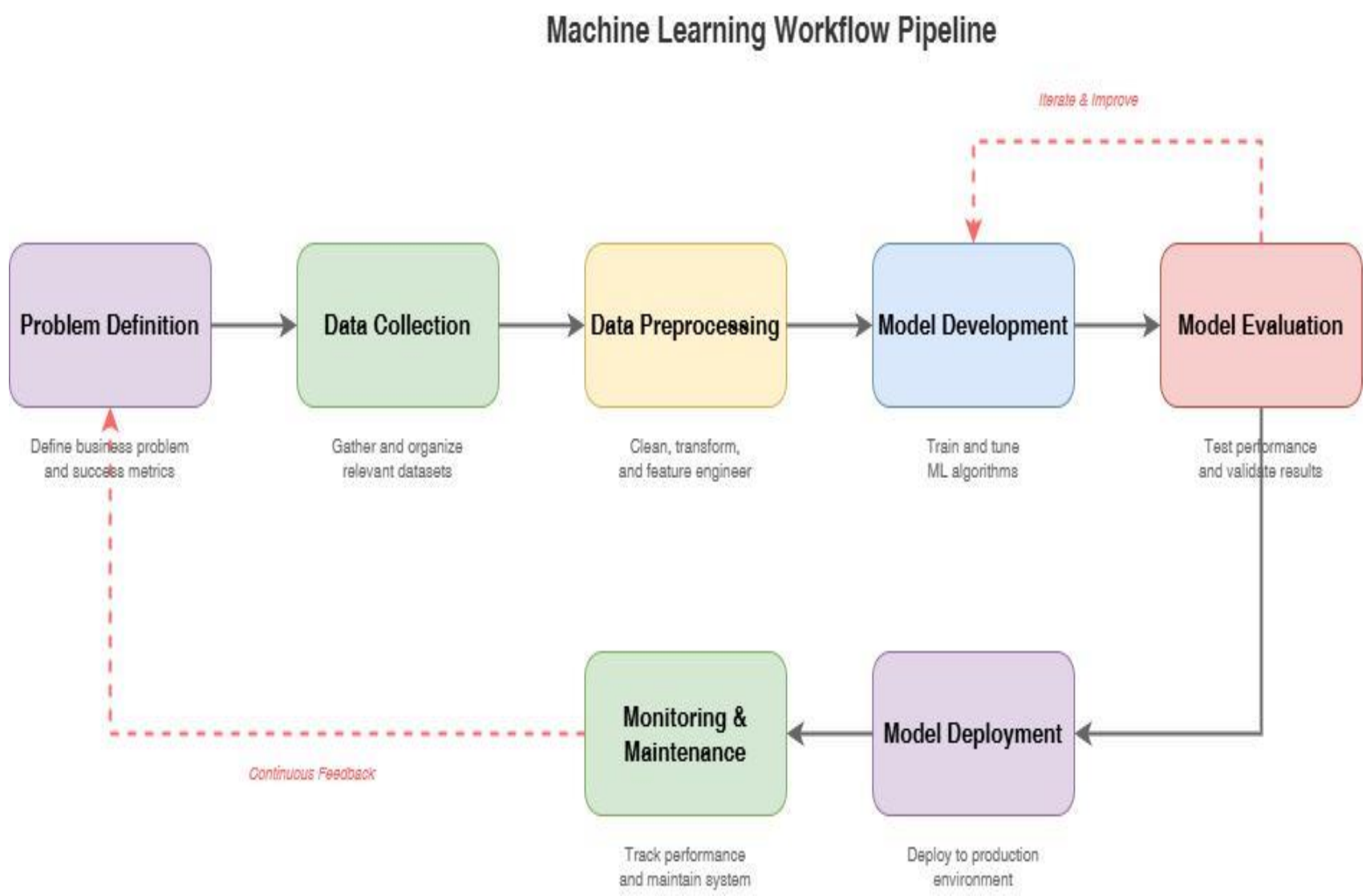
The Data Layer manages the storage and retrieval of data used in the system. It maintains structured healthcare data, which is used for model training, testing, and prediction. This layer ensures data consistency, integrity, and efficient access, acting as the foundation for all system operations.

5.2 Process Flow

The system follows a structured workflow:

1. The dataset is loaded into the system.
2. Data preprocessing is performed to clean and prepare the dataset.
3. Missing values are handled, and categorical variables are encoded.
4. The dataset is split into training and testing sets.
5. The machine learning model is trained using training data.
6. The trained model is used to make predictions on test data.
7. Performance metrics such as accuracy are calculated.
8. Results are displayed to the user.

Flowchart



5.3 Information Flow

The system processes data in a sequential pipeline:

- Raw data is collected from the dataset
- Data is cleaned and transformed
- Processed data is passed to the ML model
- The model generates predictions
- Output is displayed to the user

5.4 Components Design

The system is divided into modular components:

- **Data Acquisition Module:** Loads dataset
- **Preprocessing Module:** Cleans and transforms data
- **Visualization Module:** Displays insights
- **Model Training Module:** Trains ML model
- **Prediction Module:** Generates predictions
- **Evaluation Module:** Calculates performance

Each component operates independently, ensuring maintainability.

5.5 Key Design Considerations

- Data quality and consistency
- Model interpretability
- Performance efficiency
- Scalability for future enhancements
- Modular system design

5.6 API Catalogue

Endpoint	Method	Description
/load_data	GET	Loads dataset
/train_model	POST	Trains model
/predict	POST	Predicts stroke
/evaluate	GET	Returns accuracy

6. Data Design

6.1 Data Model

The dataset consists of structured healthcare attributes used for prediction.

Independent Variables:

- Age
- BMI
- Glucose Level
- Hypertension
- Heart Disease
- Smoking Status

Dependent Variable:

- Stroke

6.2 Data Access Mechanism

Data is accessed using the Pandas library, enabling efficient manipulation and analysis.

```
df = pd.read_csv("healthcare-dataset-stroke-data.csv")
```

6.3 Data Retention Policies

- Data is stored locally

- Used for training and testing purposes
- No long-term retention policy implemented

6.4 Data Migration

- Dataset is manually uploaded
- No automated data migration process

7. Interfaces

The system provides a Jupyter Notebook-based interface that allows users to:

- Visualize data using graphs
- View model predictions
- Analyze performance metrics

8. State and Session Management

The system operates in a stateless manner:

- Each execution is independent
- No session data is stored
- Results are generated dynamically

9. Caching

- The trained model is stored in memory
- Reduces repeated computation
- Improves system performance

10. Non-Functional Requirements

- Reliability
- Maintainability
- Usability
- Scalability
- Accuracy

10.1 Security Aspects

- Data is stored locally
- No external API exposure
- Ensures data confidentiality

10.2 Performance Aspects

- Efficient data processing using Pandas
- Fast model training
- Quick prediction response

11. References

- Python Documentation
- Scikit-learn Documentation
- Stroke Dataset